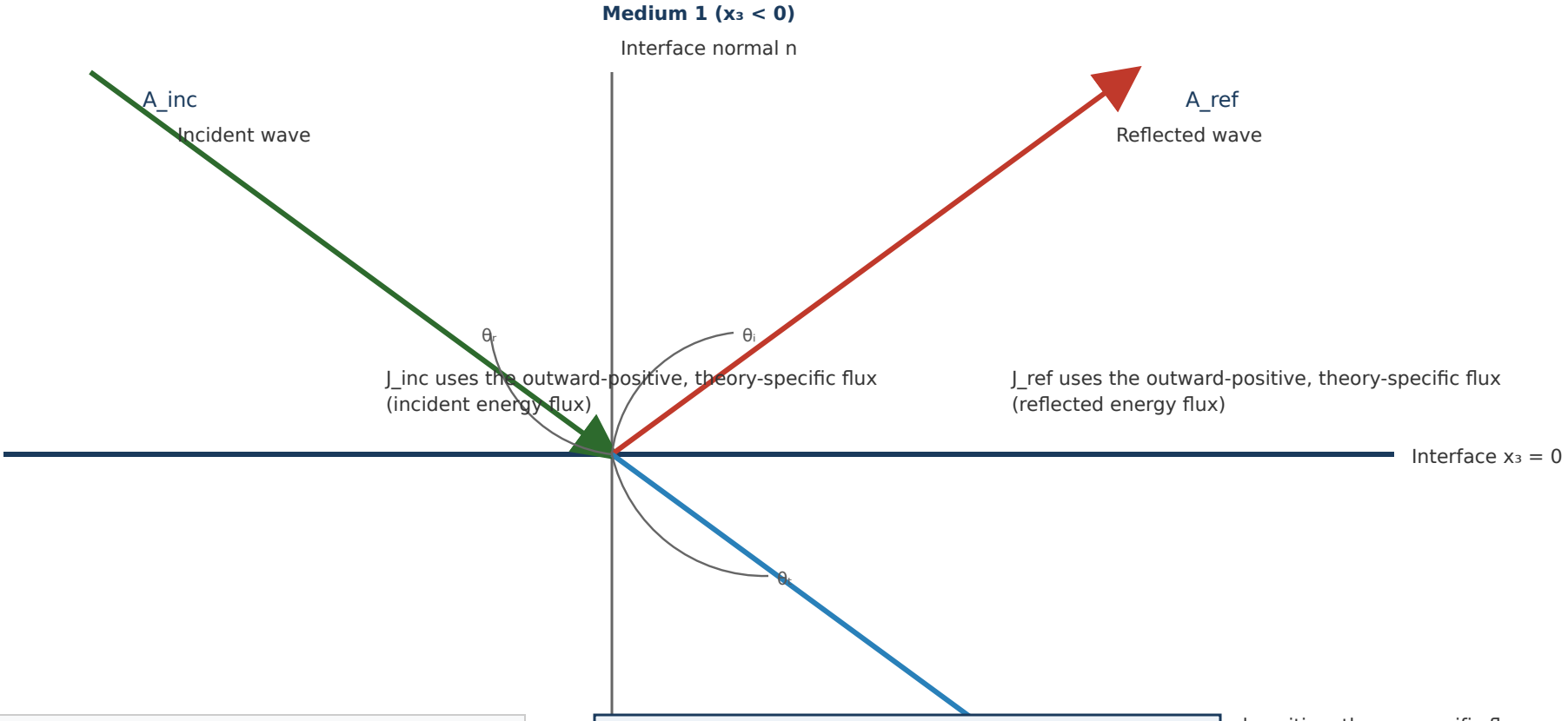
Interface reflection and transmission

Incident, reflected, and transmitted waves at a bonded interface between two media



Interface conditions (perfect bonding):

$u^{(1)}_i = u^{(2)}_i$ (displacement continuity)

$\sigma^{(1)}_{ji} n_j = \sigma^{(2)}_{ji} n_j$ (traction equilibrium)

$T^{(1)} = T^{(2)}$ (temperature continuity)

$q^{(1)}_i n_i = q^{(2)}_i n_i$ (heat flux continuity)

Energy conservation:

$R + T + A_{int} = 1$

R: reflected energy fraction

T: transmitted energy fraction

A_int: interface-local residual (not bulk loss)

total-field fluxes include all modes and cross terms

ard-positive, theory-specific flux
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For generalized theories, en
Benchmark status is not inde
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d wave